Probabilistic Retrieval Models

Review of Previous Models

- **Example Query:** machine learning

Model	**Output**
Boolean	Relevant / Not Relevant
Vector Space	Similarity scores (0.83, 0.71, 0.30)
Probabilistic	Probability of relevance (0.92, 0.74, 0.28)

- Difference is **how relevance is interpreted**

Three Classes of IR Models

- **Document D1:** “machine learning for data mining”
- **Document D2:** “introduction to databases”

Model	**Behavior**
Boolean	D1 matches, D2 does not
Vector Space	D1 is closer to the query
Probabilistic	D1 is more likely to be relevant

Limitations of Boolean Model (Example)

- **Query:** machine AND learning

Document	Content	Result
D1	Deep discussion of ML	Relevant
D2	Mentions ML briefly	Relevant

- Boolean model **cannot distinguish importance**

Limitations of Vector Space Model (Example)

- **Query:** machine learning

- D1: “machine” appears 20 times, “learning” 0 times
- D2: “machine” 5 times, “learning” 5 times

=> Cosine similarity may favor **D1**, but users usually prefer **D2**

Motivation for Probabilistic Models

- New question:
  - “Which document is **most likely** to be relevant?”
- Not:
  - “How similar is it?”
- But:
  - **“How confident are we that it is relevant?”**

The Probabilistic Retrieval Problem

- For each document *d* and query *q*:

$$P(R=1|d,q)$$

  - Where:
    - R = 1 → relevant
    - R = 0 → not relevant

Probability Ranking Principle (PRP)

- **Principle**
  - Ranking documents by decreasing probability of relevance
    gives optimal retrieval effectiveness.
- **Example**
  - D1: 0.85
  - D2: 0.60
  - D3: 0.20

=> Optimal ranking: D1 → D2 → D3

Intuitive Comparison

Query: quantum mechanics

- Boolean: returns many documents
- Vector Space: ranks by similarity
- Probabilistic: ranks by **confidence of relevance**

=> Closest to human reasoning

Binary Independence Model (BIM)

- **Assumptions**
  - Terms are either present or absent
  - Terms are independent
- **Example**

  Query: quantum mechanics
  - D1: quantum T, mechanics T
  - D2: quantum T, mechanics F

  => Probability(D1 relevant) > Probability(D2 relevant)

Binary Representation (Example)

Term	D1	D2
quantum	1	1
mechanics	1	0

No term frequency
Only presence / absence

What Makes a Term Important?

Term	Observation
email	Appears everywhere
spam	Appears mainly in spam-related documents

“spam” is much more informative

Connection to IDF (Intuition)

- High IDF ⇔
  - High P(term | relevant)
  - Low P(term | non-relevant)
- IDF has a **probabilistic interpretation**

Limitations of BIM (Example)

- D1: “spam spam spam spam email”
- D2: “spam email”

=>BIM treats D1 and D2 almost the same. No use of term frequency

⇒BM25 improves probabilistic models by:

  – Using term frequency
  – Preventing TF from growing indefinitely

  => More realistic ranking

What Does BM25 Consider?

- BM25 score depends on:
  - Term frequency
  - Document frequency
  - Document length
  - Tunable parameters (k_1, b)

=> Designed for real-world text

TF Saturation (Example)

- TF = 1 → big impact
- TF = 10 → smaller impact
- TF = 100 → almost no extra benefit

=> Avoids keyword stuffing

BM25 vs TF–IDF (Example)

- Long document (10 pages):
- TF–IDF: often favored
- BM25: **penalized for length**

=> BM25 is more stable

Practical Ranking Example

- **Query:** machine learning

Rank	TF–IDF	BM25
1	Doc A	Doc B
2	Doc B	Doc A

- BM25 rankings often align better with users

Relevance Feedback (Example)

- User feedback:
  - D1: Relevant
  - D2: Irrelevant
- Probabilistic model:
  - Updates term probabilities
  - Learns from user behavior

=> Boolean model cannot do this

Probabilistic Models in Practice

- M25 is used in:
  - Lucene / Elasticsearch
  - Academic search engines
  - As a baseline for neural IR models

=> Still highly relevant today

Steps in Probabilistic Retrieval Models

- **Modeling Relevance Probability**
  - Compute the probability that a document is relevant to a query.
- **Probabilistic Modeling Using Bayes' Theorem**
  - Apply Bayes' Theorem to estimate $P(D|Q)$, the probability that document D is relevant given query Q.
- **Calculating Relevance Probability**
  - Use the model to calculate the relevance probability for each document based on the query.
- **Ranking Documents**
  - Rank the documents based on their relevance probability, with the highest probability being ranked first.
- **Relevance Feedback and Improvement**
  - Adjust the model based on user feedback to improve future retrieval accuracy.

Step 1 – Modeling Relevance Probability

- **Objective**: Compute the probability of a document being relevant given a query.
- **Formula**: $P(D|Q)$, where:
  - D = Document
  - Q = Query
- The goal is to determine the likelihood that a document is relevant to the query. This is a probabilistic approach rather than a similarity-based one.

Probabilistic Modeling Using Bayes' Theorem

- **Bayes' Theorem** helps in calculating the probability of a document being relevant to a query:

$$P(D|Q) = \frac{P(Q|D) \cdot P(D)}{P(Q)}$$

- Where:
  - P(Q|D) = Likelihood of observing the query Q given document D.
  - P(D) = Prior probability that document D is relevant.
  - P(Q) = Probability of observing query Q (usually constant across documents)

Step 3 – Calculating Relevance Probability

- **Objective**: Apply the probabilistic model to compute the relevance probability for each document.
- **Process**:
  - Calculate the likelihood of relevance for each document given the query using the formula from Step 2.
  - Use this model to estimate the probability that a document is relevant.
- The model helps compute the relevance probability for each document in the corpus.

Step 4 – Ranking Documents

- **Objective**: Rank the documents based on their calculated relevance probabilities.
- **Ranking Process**:
  - Documents are sorted by their relevance probability $P(D|Q)$).
  - The higher the probability, the more relevant the document is to the query.
- The ranking is based on the likelihood that each document is relevant to the given query.

Step 5 – Relevance Feedback & Improvement

- **Objective**: Adjust the model based on user feedback.
- **Process**:
  - **Relevance Feedback**: If the user indicates which documents are relevant or not, update the model.
  - **Improvement**: Modify the weights or probabilities associated with terms based on user feedback to improve the accuracy of future retrieval results.
- Relevance feedback allows for adaptive learning, making the model smarter as it receives more feedback.

Manual Probabilistic Retrieval Models

Given the document collection:

DocID	Content
D1	machine learning is fun
D2	deep learning for image recognition
D3	machine learning and data mining
D4	data mining techniques

Tasks: *Query: (machine AND learning) AND NOT deep*

1. *Modeling Relevance Probability*
2. *Probabilistic Modeling*
3. *Document Probability Calculation*
4. *Ranking Documents*
5. *Feedback and Improvement*

Summary

Model	**Core Idea**
Boolean	Filtering
Vector Space	Similarity
Probabilistic	Likelihood of relevance